PROJECT PROPOSAL

AI Smart Study Planner

An AI Powered Weekly Study Schedule Generator for University Students

Course	CSE 4104 – Software Development III
Team Name	CSE4104-7E-T06
Section	7E
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1. Project Title

AI Smart Study Planner: An AI Powered Weekly Study Schedule Generator for University Students

2. Problem Statement

What problem exists : University students handle several courses at the same time, each with its own assignments, quizzes, and exams. Deadlines are scattered across notices, messages and notebooks. Students then have to decide, on their own, what to study, when to study it and how long to spend on each subject. Many end up studying at the last minute, giving too much time to favorite subjects and neglecting harder ones until exams are close.

Why it is important : Poor time planning leads to missed deadlines, stress, rushed preparation and lower academic performance. A student's available time is limited, so how that time is distributed matters as much as how much time there is.

Why existing solutions are insufficient : Common tools such as paper planners, calendar apps and generic to-do lists only record tasks. They do not consider subject difficulty, estimated work hours or how many hours a student can actually study each day. They also do not rebalance the workload when deadlines cluster together and they give no explanation for why a task is placed on a certain day.

Why AI is relevant : Building a good schedule means weighing many factors together. Deadline urgency, subject difficulty, estimated effort and daily time limits. Writing fixed rules for every possible student situation is impractical. A large language model such as Google Gemini can reason over this mixed information produce a balanced 7 day plan and explain its suggestions in plain language. This turns a passive task list into an active planning assistant.

3. Goals

- Allow students to organize their subjects, assignments and exam dates in one place.
- Generate personalized 7 day study schedules using Google Gemini AI based on deadlines, difficulty, estimated hours and daily available study time.
- Provide short AI explanations so students understand why the schedule is arranged the way it is.
- Help students track progress through a clear academic dashboard and study session completion tracking.
- Let students save, review and reuse past study plans.
- Deliver a simple, minimal, university focused web application that improves student productivity.

4. Proposed Features

Feature	Description
Authentication	Secure registration and login so each student has a private account (password hashing and token-based sessions).
Subject Management	Add, edit, and delete subjects with a difficulty level (Easy / Medium / Hard).
Assignment & Exam Management	Create assignments and exams linked to a subject, with deadline, estimated work hours and status.
Academic Progress Dashboard	Overview of upcoming deadlines, completed vs pending tasks and study progress per subject.
AI Weekly Study Plan Generator	Student enters available study hours per day. The app sends the academic data to Gemini and receives a structured 7 day plan with daily study sessions.
AI Explanations	Each generated plan includes short reasons for the choices (why a hard subject with a near deadline gets more time).

Feature	Description
Task & Session Tracking	Mark tasks and study sessions as completed. Progress updates on the dashboard.
Saved Study Plans	Save generated plans, view past plans and delete or regenerate them.

5. Technology Stack

Layer	Technology
Frontend	React.js, CSS
Backend	Node.js + Express.js
Database	MongoDB
AI Integration	Google Gemini API (called from the backend so the API key stays secret)
Deployment	Vercel (frontend) / Render (backend)
Tools	Git & GitHub, VS Code

How AI is integrated : The React frontend sends a request to the Express backend. The backend collects the student's subjects, deadlines, estimated hours and daily available time from MongoDB, builds a structured prompt and sends it to the Gemini API. Gemini returns a 7 day plan in a structured (JSON) format with explanations. The backend validates the response, stores the plan in MongoDB and the frontend displays it as a weekly schedule.

6. Expected Outcome

What users gain : Students get one place to manage academic work and a ready made, personalized weekly plan that tells them what to study and when, along with a clear view of deadlines and progress.

How it improves the existing workflow : Instead of manually planning each week using scattered notes, students enter their information once and receive a balanced schedule in seconds. Planning time is reduced and the risk of missing deadlines or leaving hard subjects until the end goes down.

Benefits of AI integration : The AI weighs deadlines, difficulty, estimated workload and available time together, adapts to each student's situation, and explains its reasoning. This makes the planner more useful than a basic to-do list or calendar.

Final deliverable : A functional AI powered web application where students can manage academic tasks, view deadlines, track progress and generate and save personalized weekly study schedules.

7. Initial Project Planning

Phase	Planned Work
Phase 1 – Planning & Setup	Proposal, GitHub repository, project structure, database design, UI sketches
Phase 2 – Core Backend & Auth	User authentication, subject / assignment / exam APIs, MongoDB models
Phase 3 – Frontend Basics	Login, subject and task pages, dashboard layout
Phase 4 – AI Integration	Gemini prompt design, plan generation API, plan display, explanations
Phase 5 – Tracking & Saved Plans	Completion tracking, progress dashboard, saved plan management
Phase 6 – Testing & Deployment	Testing, bug fixing, deployment, documentation, final presentation

Team collaboration : The team will use a shared GitHub repository with separate branches per feature and regular progress meetings. Work will be divided across frontend, backend/database and AI integration with all members contributing to testing and documentation.